ARSALAN AMIR ALI

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EDUCATION

York University — Lassonde School of Engineering Sep 2024 – Apr 2029 (Expected)

BSc Computer Science (Specialized Honours), Specialization in Artificial Intelligence

Relevant coursework: Data Structures · Operating Systems · Database Systems · Computer Organization · Theory of Computation · Advanced OOP · Probability · Linear Algebra

Awards: International Entrance Award for Academic Excellence · Lassonde Entrance Scholarship

Beaconhouse School System Sep 2022 – Jul 2024

IB Diploma Programme, Computer Science · Habib University Meritorious Award (top 10 students) · 75% merit scholarship

TECHNICAL SKILLS

Languages: Python · Java · C · JavaScript · SQL

AI / Machine Learning: LLM APIs (Claude, OpenAI, Azure OpenAI, AWS Bedrock) · Retrieval-Augmented Generation (RAG) · AI Agents (LangGraph) · Model Context Protocol (MCP) · LLM Evaluation · LLM Observability (Langfuse) · Prompt Engineering · Vector Search (pgvector) · PyTorch · Fine-Tuning (LoRA/QLoRA, Unsloth) · Triton GPU Kernels · Artificial Intelligence

Backend & Data: FastAPI · Node.js · Express · REST APIs · API Integration · PostgreSQL · Supabase · Redis · Data Pipelines

Frontend & DevOps: Next.js · React · Docker · Git/GitHub · Linux · CI/CD (GitHub Actions) · Railway · Software Testing

Certifications: Anthropic Academy — Claude API · Model Context Protocol (Intro + Advanced)

EXPERIENCE & PROJECTS

ML Systems Engineer (Independent Project) — KernelForge Jul 2026 – Present

Python · PyTorch · Triton · LoRA/QLoRA (Unsloth) · github.com/Arsalan-05/KernelForge

- Built a GPU-validated verification harness that automatically catches incorrect LLM-generated Triton kernels — correctness checks against PyTorch eager execution, speedup benchmarking, and sandboxed timeout isolation for untrusted generated code
- Fine-tuning an LLM (LoRA/QLoRA) to generate and optimize Triton GPU kernels for high-frequency LLM-inference operations (quantized matmul, attention variants, KV-cache ops, normalization fusion)
- Building a verification-gated dataset and fine-tuning pipeline positioned as a data-efficient, explainable alternative to broad-coverage systems (KernelLLM, Kevin-32B)

Founder & AI Software Engineer (Independent Project) — FinSight AI May 2026 – Present

Python · FastAPI · LangGraph · MCP · PostgreSQL + pgvector · Azure OpenAI · AWS Bedrock · Langfuse · Next.js · Docker · Railway · github.com/Arsalan-05/FinSight_AI

- Shipped a production RAG finance agent (v2.0) solo, grounding spending analysis in a knowledge base built over transaction history (chunking, embeddings, pgvector retrieval)
- Built a CI-gated automated eval suite (retrieval recall@k and answer-quality metrics) that catches regressions before deploy, plus end-to-end tracing of every LLM and tool call (latency, tokens, cost per request)
- Reduced per-request cost and latency by routing across Azure OpenAI and AWS Bedrock with response caching; hardened the agent with step/cost limits, retries, timeouts, fallbacks, and automated prompt-injection guardrail tests
- Deployed and maintain the full production stack (Next.js frontend, FastAPI backend, PostgreSQL + pgvector) on Railway with Docker and GitHub Actions CI

Co-Founder & AI Software Engineer (Co-Built Project) — RoomFinderAI May 2025 – Present

Node.js · Express · Supabase · PostgreSQL · OpenAI · Cloudflare Workers AI · Stripe · Railway · roomfinderai.com

- Co-built and shipped an AI rental search, negotiation, and marketplace platform now live on web, iOS (App Store), and Android (Google Play): 60+ REST endpoints, 27 SQL migrations, real-time chat via Supabase Realtime
- Designed an LLM negotiation engine with an 8-phase conversation flow, landlord personality modeling, and a learning loop that improves negotiation templates from live usage data
- Built a true-cost scoring engine fusing 15+ external data sources (RentCast, Google Distance Matrix, GreatSchools, FEMA, BLS) and a vision model (Llama 3.2 Vision) that grades rental units A+ to C- in real time
- Owned production auth (email, Google OAuth, Apple Sign-In), Stripe payments, and identity verification end-to-end

INVOLVEMENT

York University: CS Hub · AWS Cloud Club · Data Science Club · AI & Data Science Club · YU Blueprint

Volunteer: AV Crew – Hackathon 3.0 · Event Coordinator – Sports Gala · Technical Crew – Drama Festival · Food Stall Coordinator – Flood Relief Carnival